
Real Neutral News (RNN)*

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Motivation



- What news do you trust?
- How can you tell the difference between fake news and real news?
- How do you know if you're getting just the facts or being persuaded by political agendas?

Text Analysis Background

Text analysis is done with all different types of modeling:

- classification, regression, supervised learning, unsupervised learning

And used for a wide variety of tasks:

- bias detection, content monitoring, summarization, and etc.

Text Analysis Background

Old School - Before AI, we hard coded text analysis which mostly involved searching text for patterns.

Classical Machine Learning Approach - Bag of Words, TF-IDF (weighted term frequency), Classification or Regression Model (Logistic Regression, Naive Bayes, SVM).

Big Step in Sentiment Analysis - Word Embeddings (Word-2-Vec) moving beyond term frequency.

Sequence Models - RNNs - process text in the order it's written.

BERT - neural network architecture to improve upon previous word embeddings.

Data Acquisition

Fake News Data

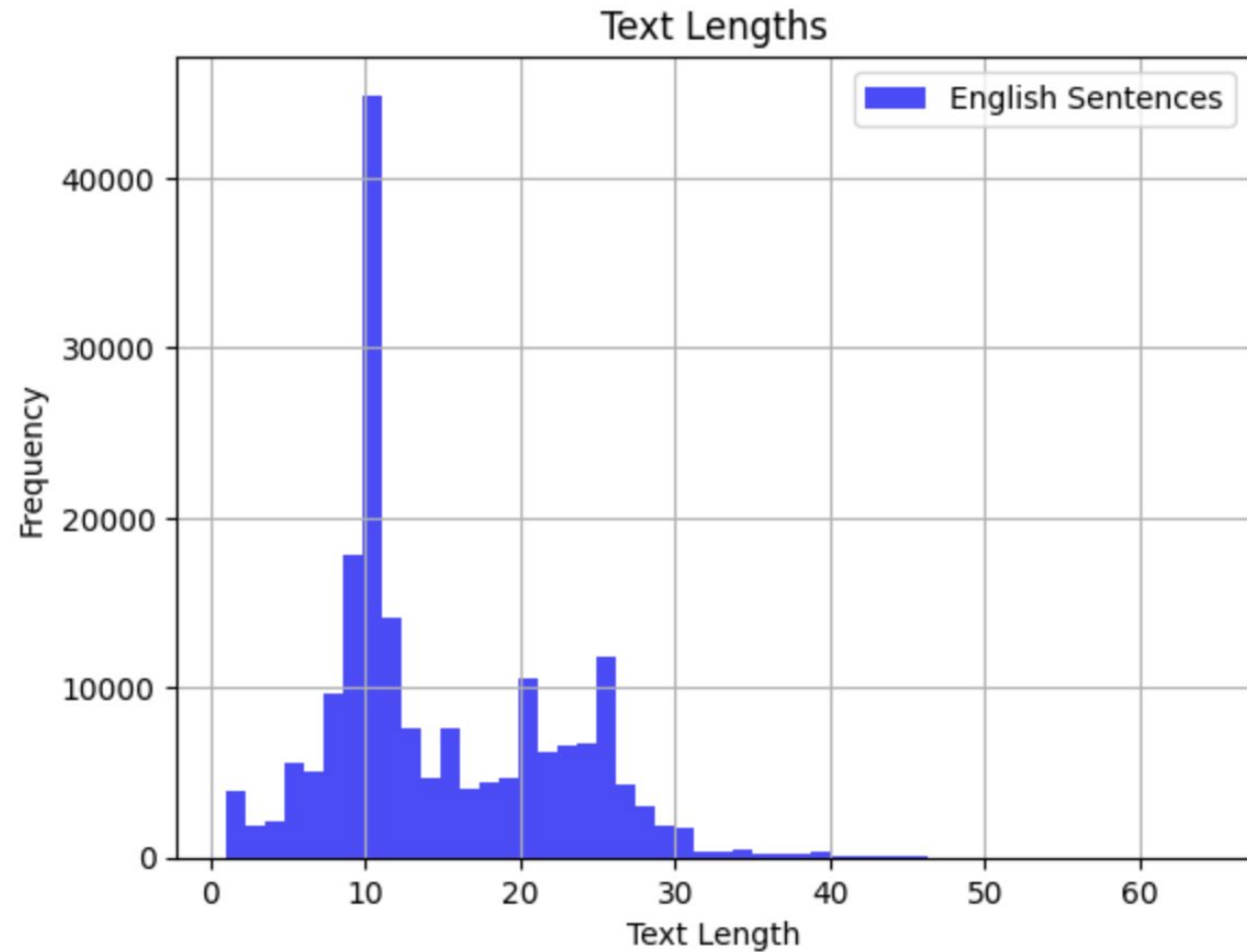
- Kaggle Dataset - ISOT Fake News Dataset (~ 44K)
 - Picked for most consistent labeling method

Political Bias Data

- Labeling by Source - Not effective
- Statements/Twitter Posts - Highest training accuracy
 - combination of 3 datasets
- Kaggle Dataset with longer news articles - Better generalization
 - (in progress)

Data Exploration

- Text Lengths
- Especially Important for



Real News Project: Data Exploration

Political Bias Data

- Word Importance

Word Cloud for Conservative Articles



Word Cloud for Liberal Articles



Data Preprocessing

- Labels 0 (fake/ left) and 1 (true/ right)
- Text as feature
- Text cleaning
 - e.g. Lowercase, punctuation and etc.
- Removed stopwords and custom ones
- Lemmatization
- Vectorization (TF-IDF and CountVectorizer)

Model Selection

Political Bias Model

- **Tested 3 Models** - Random Forest, SVM and Logistic Regression
 - Random Forest - extremely slow and poor results
 - SVM and Logistic Regression had almost identical results.
 - Grid Search for optimal parameters.

```
log_reg = LogisticRegression(  
    solver = "lbfgs",  
    penalty= "l2",  
    dual = False,  
    max_iter = 2000,  
    tol= .000175,  
    random_state=42  
)
```

Results: Reliable News

Split: 70/30 ~ 39K total obs.

	Logistic Regression
Accuracy	0.96
Precision	0.96
Recall	0.96
F1 Score	0.96

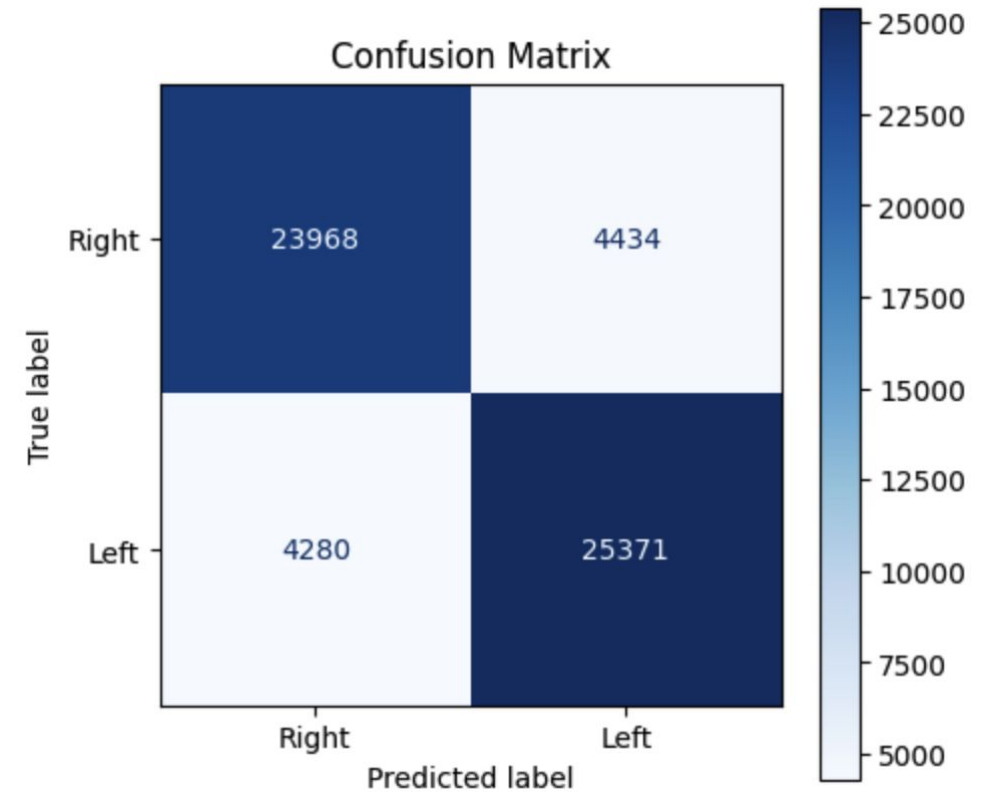
Results: Political Bias

Accuracy: .85

Precision: .85

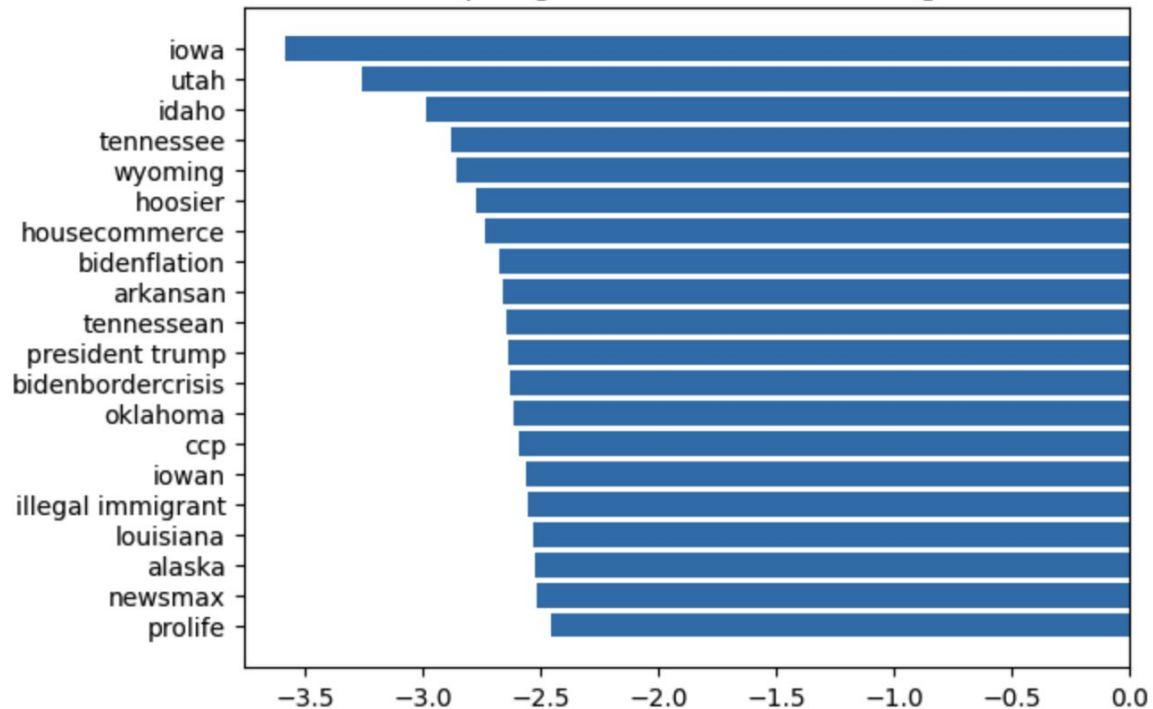
Recall: .85

f1 Score: .85

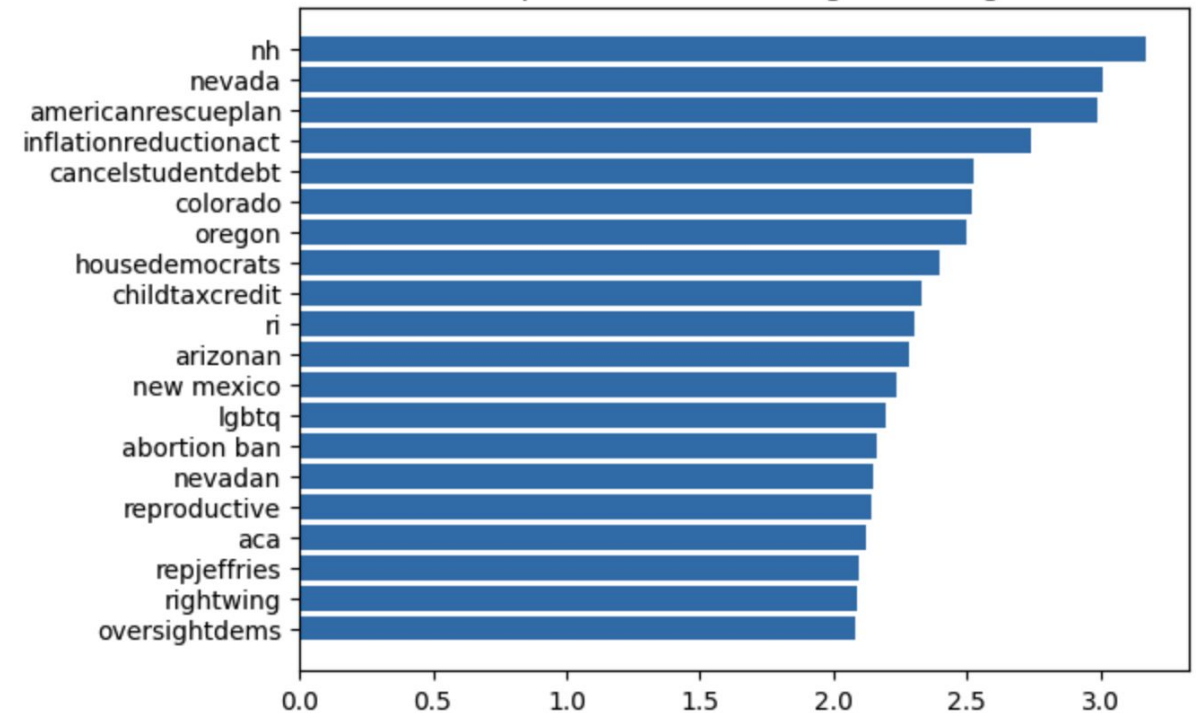


Results: Political Bias

Top Negative Words - Left Leaning



Top Positive Words - Right Leaning



Real News Project: User Interface

Simple User Interface

- Used Gradio for the interface
- User inputs their first name and copies text into the box.
- Output real/fake, left/right and probability score
 - Also outputs top features for political bias score

Time for Live Demo!!!!

Limitations

- Lack of generalization for unseen data (outside of test set)
 - Potential cause - timing of the datasets (e.g. 2016 news vs 2026 news)
- Vectorization methods not understanding context
- Ethical concern: jeopardizing users' trust
 - i.e. stating article is real when fake
 - i.e. stating an article is 100% left/right leaning when its not

Future Work

- Try TF-IDF for bias detection model, fine tune stop words, try a neural network or BERT model to capture word sequence.
- Trouble shoot generalization. The problem is that news is ever changing. Our model would need to constantly retrain on new events.
- Further analysis of misses.



Thank you for Listening!

Any Questions?